Ex:No: 3 **Date:** 1/04/26	**Comprehensive Statistical Analysis of Retail Sales Data**

Problem Description:

A retail organization maintains sales data for 5 products across 12 months. Each product has:

- **Numerical attributes:** MonthlySales, Price
- **Categorical attributes:** ProductName, Category, Region

Due to business operations, the dataset may contain variability in sales and customer regions. The organization wants to:

1. Analyze **frequency distributions** of products and regions.
2. Calculate **measures of central tendency** and **dispersion** for sales data.
3. Conduct **hypothesis testing** to compare sales across categories and regions.
4. Model data using **statistical distributions** (Binomial, Poisson, Normal, Chi-squared) for probability analysis.

Objective:

To perform **complete statistical analysis** on retail sales data in R by:

- Exploring data structure and identifying trends
- Calculating **mean, median, mode, variance, SD, range**
- Performing **T-Test, ANOVA, F-Test, Mann-Whitney U-Test, Fisher's Exact Test, Kruskal-Wallis Test, Bartlett's Test**
- Applying **Binomial, Poisson, and Normal distributions**

This provides a **framework for business insights and decision-making**.

Procedure:

- **Step 1: Create and Explore Dataset**

```
# Sample retail sales dataset with ProductID, ProductName, Category, Region,
# MonthlySales and Price

sales_data <- data.frame(
 ProductID = 1:8,
 ProductName = c("Laptop", "Mobile", "Tablet", "Camera", "Headphone",
"Mouse", "Keyboard", "Charger"),
 Category = c("Electronics", "Electronics", "Electronics", "Electronics",
"Accessories", "Accessories", "Accessories", "Accessories"),
 Region = c("North", "South", "East", "West", "North", "South", "East",
"West"),
 MonthlySales = c(50, 120, 80, 30, 150, 70, 60, 55),
 Price = c(700, 300, 250, 400, 50, 20, 30, 25)
)
```

"data.frame": 8 obs of 6 Variables:

$ product Id : int 1 2 3 4 5 6 7 8

$ category : chr "Electronics" "Electronics" "Electronics" - - - -

$ Region : chr "North" "South" "East" "West" . . .

$ monthly sales: num 50 120 80 30 150 70 60 55

$ price : num 700 300 250 400 50 20 30 2 5

Product ID	Product name	Category	Region
min: 1.00	length: 8	length: 8	length: 8
1st Qu: 2.75	class: character	class: character	class: character
median: 4.50	mode: characters	mode: character	mode: characters.
Mean: 4.50			
3rd Qu: 6.25			
max: 8.00			

Monthlysales	Price
min: 30.00	min: 10.00
1st Qu: 53.75	1st Qu: 28.75
median: 65.00	median: 150.00
mean: 76.88	mean: 221.88
3rd Qu: 90.00	3rd Qu: 325.00
max: 150.00	max: 700.00

mean: 76.875

median: 65

mode 30 50 55 60 70 80 120 150

Accessories	Electronics
4	4

East	North	South	West
2	2	2	2

Explore data structure sales_data. Use str()

print (str (sale_data)

Summary statistics sales_data

print (summary(sales_data))

Frequency distributions Category and Region. Use table()

table (sales_data $ category)

table (salesdata $ Region)

- **Step 2: Measures of Central Tendency**

  # Mean, Median, Mode for MonthlySales

mean_sales = mean (sales_data $ monthly sales)

install. packages ("modeest") ; library (modeest)

mode sales = mfv (Sales_data $ monthlysales)

median_sales = median (sales_data $ monthlysales)

Print results mean_sales; median_sales; mode_sales

print (mean_Sales)

print (median_sales)

print (mode_sales).

Variance : 1563.834

Standard Deviation : 39.54841

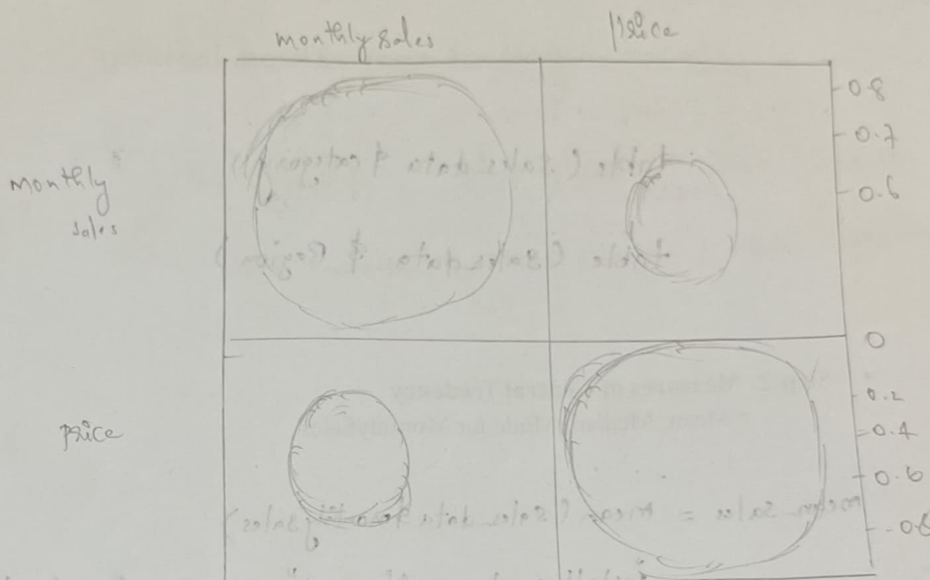
Range : 30 150

Pearson Correlation : -0.3179059

"Higher price reduce sales"

Pearson correlation : -0.3179059

>corplot



	Product ID	monthly Sales	price
Product ID	1.000	-0.1290	-0.854
monthly sales	-0.1290	1.000	-0.317
price	-0.8540	-0.3179	1.000

- **Step 3: Measures of Dispersion**

  # Variance, Standard Deviation and Range of sales for MonthlySales

```
variance_sales = Var( Sales_data $monthlysales )
sd_sales      =  Sqrt(variance_sales)        // or use sd()
max (Sales_data $monthlysales)
min (sales_data $ monthlysales)
range_sales  =  max - min                    // or use range()
```

  # Print results variance_sales; sd_sales; range_sales

```
print (variance_sales)
print ( sd_sales)
print ( ranges sales )
```

- **Step 4: Correlation Analysis for Retail Sales Data**
  - **Correlation Coefficient:** To analyze the relationship between Monthly Sales and Product Price using correlation analysis

    # Select numerical variables

```
numerical_val  <-  c ( Sales_data $monthlysales, sales_data $ price)
```

Compute Pearson correlation

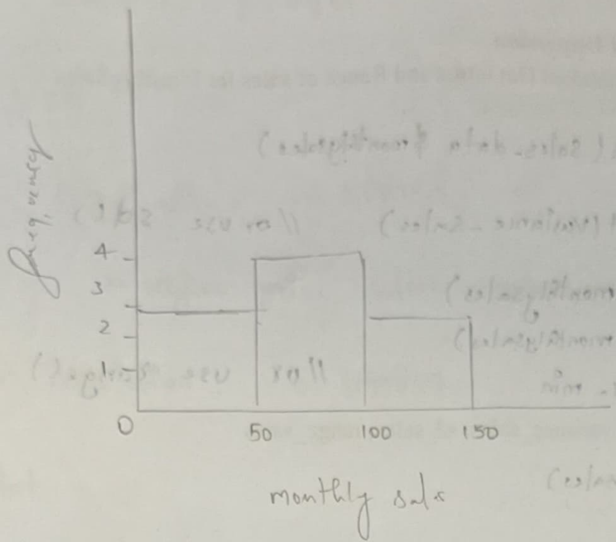
```
cor <- cor (Sales_data $monthly sales, sales_data $ price )
```

#Print correlation_value

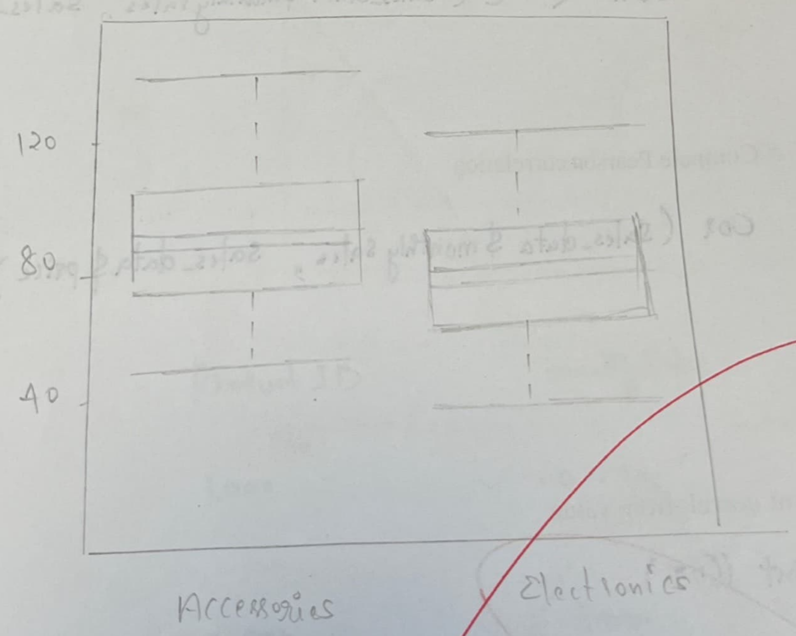
```
print (cor )
```

- A **positive value** indicates higher-priced products tend to have higher sales.
- A **negative value** indicates higher prices reduce sales.
- A value **near zero** indicates weak or no linear relationship.

Histogram plot.



Box plot



- **Correlation Matrix for Retail Sales Data:** To analyze pairwise relationships among numerical variables using a correlation matrix
  # Compute correlation matrix

  Cor-matrix <- cor (num-data)

  # Display correlation matrix

  print (cor-matrix)

- **Visualizing Correlation Matrix:**
  # Install and apply corrplot for correlation matrix ("corrplot")

  install. packages ("corrplot")

  library (corrplot)

  corrplot (cor_matrix , method = "circle")

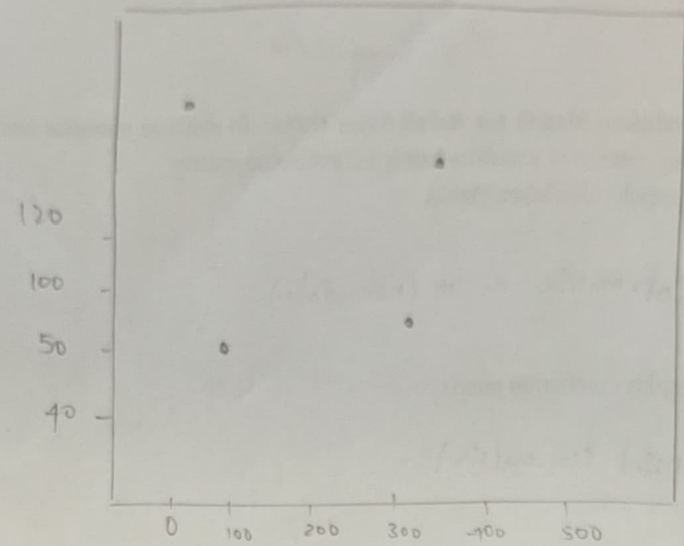
- **Step 5: Data Visualization**
  - **Histogram of Monthly Sales:** To understand the distribution and frequency pattern of monthly sales

hist

hist (sales_data $MonthlySales, main = "Histogram of monthly sales",

xlab = "Monthly Sales",

col = "light blue")



T-test:

welch Two Sample t-test

data: sales accessories and sales. electronics

t= 0.46332, df= 5.901 , p-value =0.6597

alternative hypothesis: true difference in means is not equal to 0, 95 percent confidence interval:

-59.16399 86.66347

Sample estimate:

mean in group accessories mean in group electronics

83.75 70.00

box plot

- **Boxplot of Monthly Sales by Category:** To compare sales distribution between Electronics and Accessories and identify outliers

boxplot (Monthly-sales ~ Category ,
Independent Dependent

boxplot (monthly-sales ~ category , data = sales-data,
main = "Sales by category",
col = c("orange", "green"))

- **Scatter Plot for Correlation Visualization:** To visually confirm the correlation between price and sales

plot (sales-data $price, sales-data $monthlySales ,
main = "price Vs Sales",
xlab = "price"
ylab = "Monthly Sales".
pch = 19)

- **Step 6: Hypothesis Testing**
  - **T-Test:** Compare Accessories vs Electronics: t.test( )

t.test (electronics , accessories)

>anova

	Df	Sum	Sq	F.value	Pr(>F)
Region	3	4184	1395	0.825	0.545
Residuals	4	6763	1691		

> F. test

F test to compare two variances

data: monthly sales by category

F = 1.2976 , num df = 3, denom df = 3 p.value = 0.8356

alternative hypothesis: true ratio of variance is not equal to 1

95 percent confidence Interval:

0. 08404294 20.03317822

Sample Estimate:

Ratio of variances

1.297554

> wilcoxon

data: monthly sales by category

w = 10, p-value = 0.6657

alternative hypothesis; true location shift is not equal to 0

> Fishers

data: ~~cont table~~ monthly sales by category.

p-value = 1

alternative hypothesis : two sided

> Kruskal wallis

data: monthly sales by region

- ANOVA: Compare sales across regions: aov()

anova_result <- aov(MonthlySales ~ Regions, data = Sales_data)

Summary (anova_result)

- **F-Test:** Compare Accessories vs Electronics: Var.test()

Var.test (electronics, accessories)

- **Mann-Whitney U-Test:** Compare Accessories vs Electronics: wilcox.test ()

Wilcox.test (electronics, accessories)

- **Fisher's Exact Test:** Category and Region fisher.test()

first form contingency_table for that use table()

table_cat_region <- table (sales_data $ category, Sales_data $ Region)

fisher.test (table_cat_region)

- **Kruskal-Wallis Test:** Compare sales across regions: Kruskal.test()

kruskal.test (MonthlySales ~ Regions, data = Sales_data)

- **Bartlett's Test:** Compare sales across regions: bartlett.test ()

bartlett.test(MonthlySales ~ Regions, data= Sales_data) kruskal-wallis chi-squared = 3 df = 3, p-value = 0.59...

Barlett test:

data: monthly sales by Region

Barlett k-squared = 2.1462, df = 3, p-value = 0.5426

Binomial Distribution:

0.2373 0.3955 0.2636 0.0879 0.0146 0.0001

Poisson Distribution

0.000458 0.00352 0.0135 0.0347 0.0667

Normal Distribution

0.0080 0.00556 0.01005 0.00499 0.0018

0.0099 0.00921 0.00865

- **Step 5: Statistical Distributions** dbinom()
  - **Binomial Distribution**
    - **Scenario:** Probability of having a "high sales" month (success = MonthlySales > 100) in 5 months

      # Identify high sales months

      ```
      high_sales <- ifelse(sales_data $monthlysales >100, 1, 0)
      ```

      # Binomial probability for 0 to 5 high-sales months

      ```
      dbinom(0:5, size =5, prob = mean (high_sales)
      ```

  - **Poisson Distribution**
    - **Scenario:** Model the probability of a number of sales events per month (λ = average MonthlySales / 10)

      ```
      lambda <- mean (sales_data $monthly sales)/10

      dpois (0:5, lambda = lambda)
      ```

  - **Normal Distribution**
    - **Scenario:** Model MonthlySales as normally distributed; find probability density for each product.

      ```
      mean_sales <-mean (sales-data $monthlysales)

      sd_ sales <- sd ( Sales_data $ monthysales)

      dnorm (sales_data $monthly sales, mean = mean_sales , sd = sd_ sales)
      ```

Result:

Thus, the program successfully demonstrates:

- **Frequency distributions** for categorical attributes (Category, Region)
- **Measures of central tendency and dispersion** for numerical attributes (MonthlySales)
- Performing **hypothesis testing** (T-Test, ANOVA, F-Test, Mann-Whitney U-Test, Fisher's Exact Test, Kruskal-Wallis Test, Bartlett's Test)
- Modeling data using **Binomial, Poisson, and Normal distributions**

This provides a **complete statistical framework for analyzing retail sales data** and supports **data-driven business decisions**.